RP1 — MASTER provenance of the symbols of Revision_Deflation

Proof, required by Hard Rule 5, that every symbol of the revision has one meaning and collides neither with the MASTER registry of the main project nor with the author's notebooks

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1. What this document proves

Hard Rule 5 of the task specification: the original files often use the same symbol for completely different objects, and the project must resolve this without ambiguity and prove that it has. The main project does so with a MASTER registry (`provenance/symbols.csv` in the folder above; provenance P1) in which every symbol has exactly one meaning.

The revision introduces new objects — rates, scale factors, a curved metric, a clock — and therefore new symbols. This document is the extension of the MASTER registry by those symbols. It proves, by six machine checks whose output is reproduced in Sec. 4:

1. no two entries of the registry of the revision share a name;
2. no new name or symbol is a name or symbol of the main registry;
3. no new name is a symbol of one of the author's fourteen notebooks (1921 symbols);
4. in every document of the revision the author's rates are written g55, g66, g77 — upright and without subscript — and every paragraph in which a component of the metric g_{55}, g_{66} or g_{77} occurs says that it is a component of the metric;
5. the names that an earlier version of the programs used for new objects although the main registry uses them for other objects no longer occur in any data file;
6. every column of every data file of the revision is a registered name or a plain descriptive word.

2. The one place where confusion was possible: the rate g55 and the component g_{55} of the metric

The author's names for the three rates are `g55`, `g66`, `g77`. The components of the metric along the same three coordinates are, in the usual notation, g_{55}, g_{66}, g_{77}. The two are different objects that occur in the same formula,

$$g_{55} = -\,\mathrm{const5}^{2}\, e^{-2\,\mathrm{g55}\,x4}.$$

The rule of the revision:

object	in documents	in programs and data files
the rate of deflation of $x5$	g55 (upright, no subscript)	`g55`
the component of the metric	g_{55} (italic, with subscript), always called a component of the metric in the same paragraph	`g_55`, or the scale factor `b_5` with $g_{55} = -b_5^2$

and likewise for $x6$ and $x7$. In a statement that holds for all three coordinates the rate is written rate_t and the scale factor b_t, $t = 5, 6, 7$. Check RS4 enforces the rule in every document.

What the notebooks use. The names `g55`, `g66`, `g77` occur in none of the fourteen notebooks. The notebooks use `g44` and `g71` as names of whole metrics (which the main project renamed, error E7 of its error log), and those of the notebooks that contain exponential scale factors write the rate as `H`. No collision arises.

3. The registry of the revision

The registry is the file `provenance/symbols_deflation.csv`, written by `code/scripts/write_symbol_registry.py`. "Origin" says whether a symbol is the author's, is taken over from the main project, or is new.

D1 Hypotheses and geometry

symbol	in programs and data files	meaning	defined in	origin
g55, g66, g77	`g55`, `g66`, `g77`	the three rates of deflation of $x5$, $x6$, $x7$ (positive constants, dimension 1/length). These are the author's names; they are written upright and never with a subscript. They are *not* components of the metric: the components are g_{55}, g_{66}, g_{77}	RP0 1; DMD 0.1; M II	author
rate_t	`rate_t`, `rate`	the rate of the coordinate xt in a statement that holds for $t = 5, 6, 7$ alike: $\text{rate}_5 = \text{g55}$, $\text{rate}_6 = \text{g66}$, $\text{rate}_7 = \text{g77}$	DMD 0.1	new
gsum	`gsum`, `gam`	$\text{g55} + \text{g66} + \text{g77}$, the total rate: the proper volume of the three deflating coordinates falls as $e^{-\text{gsum}\,x4}$ (`gam`: gsum in units of 100 km/s/Mpc)	DMD 1.1; M II	new
gsq, gprod	`gsq`, `gprod`	$\text{g55}^2 + \text{g66}^2 + \text{g77}^2$ and $\text{g55}\,\text{g66} + \text{g55}\,\text{g77} + \text{g66}\,\text{g77}$; $\text{gsum}^2 = \text{gsq} + 2\,\text{gprod}$	DMD 1.3	new
const5, const6, const7; const_t	`const5`, `const6`, `const7`, `const_t`, `cst`	the three positive constants of the author's metric (his word: const): the values of the scale factors at $x4 = 0$	RP0 1; DMD 0.1	author
b_5, b_6, b_7; b_t	`b_5`, `b_6`, `b_7`, `b_t`	the scale factors of the deflating coordinates, $b_t(x4) = \text{const}_t\, e^{-\text{rate}_t\, x4}$: a coordinate distance dxt is the proper length $b_t\,\lvert dxt\rvert$. Always written with the index 5, 6, 7 or t (the letter b without index is a Ginzburg-Landau coefficient of the main registry)	DMD 1.1; M II	new
V_d	`Vd`	$b_5 b_6 b_7 = \sqrt{\lvert\det g\rvert}$, the proper volume of a unit coordinate cell of the deflating coordinates; $V_d \propto e^{-\text{gsum}\,x4}$	DMD 1.1; RS1 1	new
g_{1644d}, g_{1671d}; $g_{\mu\nu}$	`g_1644d`, `g_1671d`	the metrics of the deflating 4+4 and 7+1 spacetimes, $\text{diag}(\eta_{00}, \dots, \eta_{44}, \eta_{55}b_5^2, \eta_{66}b_6^2, \eta_{77}b_7^2)$ with $\eta = \eta_{1644}$ or η_{1671}; in both, $g_{tt} = -\text{const}_t^2 e^{-2\,\text{rate}_t\, x4}$	DMD 1.1	author
μ, ν, ρ, σ (indices)	`mu`, `nu`, `rho`, `sigma`, `lam`	coordinate indices with the values $0, \dots, 7$ on the deflating spacetimes (only as sub- or superscripts; as stand-alone letters μ, ν, λ keep the meanings of the main registry). The capital indices A, B label the orthonormal frame: the frame vector A points along the coordinate xA	DMD 1.1	new
${e_A}^\mu$, ${e^A}_\mu$	`e_A^mu`	the orthonormal frame and its inverse: ${e^A}_\mu = \text{diag}(1, 1, 1, 1, 1, b_5, b_6, b_7)$	DMD 1.1	new
$\Gamma^\rho{}_{\mu\nu}$	`Gamma`, `christoffel`	the Christoffel symbols of the deflating metric (three indices; the letter Γ with square brackets is the effective action of the main registry)	DMD 1.2	new

symbol	in programs and data files	meaning	defined in	origin
$R^{\rho}{}_{\sigma\mu\nu}$, $R^{\mu}{}_{\nu}$, R	`riemann`, `ricci`, `Ric44`, `Ric55`, `Ric66`, `Ric77`, `Rscalar`	Riemann tensor, Ricci tensor and curvature scalar of the deflating metric	DMD 1.3	new
$\mathcal{G}^{\mu}{}_{\nu}$	`Ein00`, `Ein44`, `Ein55`, `Ein66`, `Ein77`, `einstein_mixed`	the Einstein tensor $R^{\mu}{}_{\nu} - \frac{1}{2} R\, \delta^{\mu}_{\nu}$ (script letter, so that it cannot be taken for Newton's constant or for a constant of the Fermi type)	DMD 1.4	new
$\mathcal{X}$; $\mathcal{X}_{\text{scale}}$, $\mathcal{X}_{(s)}$	`killing_vectors`, `killing_count`	a Killing vector field of the deflating metric (script letter; the plain letter X is the set of coordinates of the main registry); the scaling vector $\partial_4 + \sum_t \text{rate}_t\, xt\, \partial_t$; the three special Killing vectors of the case of equal rates, $s = 5, 6, 7$	DMD 1.5, 1.6	new
$\bar{x}4$; $\bar{x}5$, $\bar{x}6$, $\bar{x}7$	—	coordinates of the case of equal rates in which the line element of $x4, \ldots, x7$ is a multiple of a flat one: $\bar{x}4 = e^{\text{g55}\, x4}/\text{g55}$, $\bar{x}t = \text{const}_t\, xt$	DMD 1.6	new
$\mathcal{H}^{\mu}{}_{\nu}$; α_{GB}, Λ_8	`Lanczos`, `alpha_GB`, `Lambda_8`	the Lanczos tensor of the Gauss-Bonnet term (script letter; H is the Hubble rate); the coupling of the Gauss-Bonnet term and the cosmological constant of Einstein-Gauss-Bonnet gravity in eight dimensions (with the index 8; Λ without index is the momentum cutoff of the main registry)	DMD 1.4	new
ς	`affine`	affine parameter along a geodesic; for a massive particle on the 7+1 spacetime, the proper time divided by the mass	DMD 1.7	new
P_{μ}	`P_1`, `P_5`	the conserved covariant momenta of a geodesic (with an index; P without index is the generator of the main registry)	DMD 1.7	new
$x4_{\text{turn}}$	`x4_turn`	the value of the ordinary time at which a classical particle with momentum along a deflating time turns back, and beyond which a mode that depends on a deflating time is extinguished (4+4): there $k_d = E_{\perp}$	DMD 1.7, 4.3	new

D2 Matrices, fields and their equations

symbol	in programs and data files	meaning	defined in	origin
γ^{μ}_{1644d}, γ^{μ}_{1671d}	`gamma_1644d`, `gamma_1671d`	the Dirac matrices of the deflating spacetimes in coordinate components, $\gamma^{\mu} = e_A{}^{\mu}\gamma^A$: $\gamma^t = \gamma^{A=t}/b_t$ for $t = 5, 6, 7$, the other five are the constant matrices γ^A_{1644}, γ^A_{1671} of the main project	DMD 2.2	new
$\omega_{\mu AB}$	`omega_(mu, A, B)`	the coefficients of the spin connection (three indices; ω_{1644}, ω_{1671} with a tag are the volume elements of the main registry)	DMD 2.3	new
Ω_{μ}	`Omega_mu`, `omega`	the spin connection as a matrix, $\frac{1}{4}\omega_{\mu AB}\gamma^A\gamma^B$; the covariant derivative of a spinor is $\nabla_{\mu}\psi = (\partial_{\mu} + \Omega_{\mu})\psi$	DMD 2.3	new
ψ_{1644d}, ψ_{1671d}	`dirac1644d`, `dirac1671d`	the fields dirac1644d and dirac1671d: the 16-component fields of the main project on the deflating spacetimes; $(\gamma^{\mu}\nabla_{\mu} - m)\psi_{1644d} = 0$ and $(i\gamma^{\mu}\nabla_{\mu} - m)\psi_{1671d} = 0$	DMD 3.1, 3.2; M III	new
$\hat{\psi}$	`psihat`	the weighted field $\sqrt{V_d}\,\psi$, which obeys the equation without spin connection	DMD 3.3; M III	new

symbol	in programs and data files	meaning	defined in	origin
$K(x4)$, $\mathsf{h}(x4)$	`K`, `h`	the generator K_{1644} and the Hamiltonian matrix h_{1671} of the main registry with the wavenumbers along the deflating coordinates divided by the scale factors: $\partial_4\hat{\psi} = K\hat{\psi}$ (4+4), $i\partial_4\hat{\psi} = \mathsf{h}\hat{\psi}$ (7+1)	DMD 3.4	main

D3 Solutions mode by mode

symbol	in programs and data files	meaning	defined in	origin
k_5, k_6, k_7; k_t	`k_5`, `k_6`, `k_7`, `k_t`	the conserved wavenumbers of a mode along the deflating coordinates (covariant components of the wave vector k_A of the main project)	DMD 3.4; M IV	new
$E_\perp$	`E_perp`	the energy of a mode, or of a classical particle, without its motion along the deflating coordinates: $\sqrt{m^2 + k_0^2 + k_1^2 + k_2^2 + k_3^2}$ (with the wavenumbers or momenta along $x0, \dots, x3$)	DMD 1.7, 3.4; M IV	new
k_d	`k_d`	the physical wavenumber along the deflating coordinates, $k_d(x4) = \left[\sum_t k_t^2/b_t(x4)^2\right]^{1/2}$; it grows as the coordinates deflate	DMD 3.4; M IV	new
$\mathsf{h}_\perp$, $K_\perp$	`h_perp`, `K_perp`	the parts of h and of K that do not contain the scale factors: the operators of the modes that do not depend on the deflating coordinates, $\mathsf{h}_\perp = m\beta + \sum_a p_a\alpha^a$ with $\mathsf{h}_\perp^2 = E_\perp^2$, and $K_\perp = -m\gamma^4 + i\sum_a k_a\gamma^4\gamma^a$ with $K_\perp^2 = -E_\perp^2$	DMD 4.1	new
γ^d, α^d	—	the frame matrix along the direction of the wave vector in the deflating coordinates, $\gamma^d = k_d^{-1}\sum_t (k_t/b_t)\gamma^t$, and $\alpha^d = \gamma^4\gamma^d$; constant in time when one rate governs	DMD 4.1	new
w; f_+, f_-; Z_+, Z_-	—	auxiliary objects of the exact solutions, used only in DMD 4.2 and 4.3: a constant eigenvector of $\mathsf{h}_\perp$ or of $K_\perp$; the two coefficient functions of the ansatz; two cylinder functions whose orders differ by one	DMD 4.2, 4.3	new
u_d	`u_d`	k_d/rate_t for a mode that depends on one deflating coordinate: the argument of the Bessel functions of the exact solutions	DMD 4.1	new
n_{cr}	`n_cr`, `n_min`, `n_mean`, `n_max`, `n_exact`, `fd5`, `fd6`	the number of particles per internal state that the deflation creates in a mode that starts empty (7+1)	DMD 4.2; M IV	new
T_{defl}	—	the temperature of the thermal spectrum of the created quanta, $\mathrm{rate}_t/2\pi$ in units with $\hbar = k_B = 1$ (7+1)	DMD 4.2; M IV	new
S_{WKB}	`S_wkb`	$\int \sqrt{k_d^2 - E_\perp^2}\, dx4$ from the turning value on: exponent of the extinction of a bounded solution (4+4)	DMD 4.3; M IV	new
φ_{bos}, $\hat{\varphi}_{\mathrm{bos}}$, m_{bos}	`phi_bos`, `Shat`, `Shat2`, `m_bos`	a mode of the Vorton-Scalar (the radial mode, $m_{\mathrm{bos}} = m_s$, or the phase mode, $m_{\mathrm{bos}} = 0$), its weighted form $\sqrt{V_d}\,\varphi_{\mathrm{bos}}$, and its mass	RS1 2	new
ϖ_b	`varpi_b`	$\sqrt{k^2 + m_{\mathrm{bos}}^2 - \mathrm{gsum}^2/4}$: the frequency of the weighted boson mode that does not depend on the deflating coordinates	RS1 2	new

D4 The deflation clock and the couplings

symbol	in programs and data files	meaning	defined in	origin
τ_d	`tau_d`	the deflation clock, $\tau_d = \mathrm{gsum}\,(x4 - x4_{\mathrm{ref}})$: the number of e-foldings by which the volume V_d has shrunk since the reference time (the letter τ without the index d is the optical depth of the main registry)	RS1 1; M V	new
τ_{pair}	`tau_pair`	the clock measured from the moment at which two fermions become bound: $\ln(D_{VS}/D_c)$	RS1 4; RS4 1	new
$\hat{\Phi}_{VS}$	`Phihat`	the weighted Vorton-Scalar $\sqrt{V_d}\,\Phi_{VS}$	RS1 1	new

D5 Dynamics of the paired state

symbol	in programs and data files	meaning	defined in	origin
Ξ_d	`Xi_d`	$\mathrm{gsum}/(\lambda\,\Delta_{\mathrm{eq}})$: decides whether the paired state can follow the deflation ($\Xi_d \ll 1$: yes)	RS1 7; M VIII	new
n_k	—	occupation of the pair of states of the single-particle energy ξ_k in the shell model of the paired state (with the index k; the letter n without index is a density)	RS1 7	new
Δ_{eq}	`gap_eq`	the equilibrium gap for the pairing strength of the moment	RS1 7	new
E_{sh}	`E_sh`	half-width in energy of the shell of states around the Fermi surface in the model of the pair dynamics; unit of energy and of inverse time there	RS1 7	new
n_{ex}	`n_ex`, `n_ex_final`	density of broken pairs (in units of E_{sh} times the density of states)	RS1 7	new

D6 Cosmology and bounds

symbol	in programs and data files	meaning	defined in	origin
G_N, G_8	`G_N`, `GN_early`, `GN_star`, `G_8`	Newton's constant of ordinary space, and the gravitational coupling of the eight-dimensional Einstein theory (assumption A2): $G_N = G_8/(\text{proper volume of the extra coordinates}) \propto e^{\mathrm{gsum}\,x4}$	RS3 1; M XI	new
a_{sp}, H	—	scale factor of ordinary space (assumption A1) and its rate $H = (da_{\mathrm{sp}}/dx4)/a_{\mathrm{sp}}$; H_0, h as in the main registry. (The author's Einstein-Lovelock notebooks use the letter H for the rate in the exponent of their metric; that rate corresponds here to g55, g66, g77)	RS3 1	new
t_0	`age`, `age_Gyr`	the age of the universe in the model at hand	RS3 2	new
δ_m, f_g, σ_8	`growth`, `f_growth`, `growth_ratio`, `f_growth_ratio`, `f_sigma8_ratio`	density contrast of matter in ordinary space; its growth rate $d\ln\delta_m/d\ln a_{\mathrm{sp}}$ today; the amplitude of clustering (only ratios of σ_8 are used)	RS3 5	new
v_{EW} (main project), n_{EW}	`v_EW`, `n_EW`	the electroweak scale of the supplement S2 of the main project, and the exponent with which it would follow the coupling under the hypothesis H2: $n_{\mathrm{EW}} = d\ln v_{\mathrm{EW}}/d\ln\alpha_{VS}$	RS2 5	new

symbol	in programs and data files	meaning	defined in	origin
ϱ_c	`rho_c`	abbreviation in the formula for n_{EW} of the scale-invariant ladder equation: $\varrho_c = \frac{1}{2}\sqrt{\alpha_{VS}/\alpha_c - 1}$ (the letter ρ without index is a density)	RS2 5	new
m_p/m_e	`mu_pe`	the ratio of the proton mass to the electron mass (always written as this ratio in the documents; `mu_pe` in the statements of the programs)	RS2 5; RS3 3	new
G_{Fermi}, α_{em}	`G_Fermi`	the Fermi constant of the weak interaction and the fine-structure constant (they enter only in the discussion of bounds)	RS3 1, 3	new

D7 Symmetries and vortex lines

symbol	in programs and data files	meaning	defined in	origin
P_c	`Pc`	$i\alpha^1\alpha^2\alpha^3\chi$, the product of the five matrices of the sector that does not depend on the extra coordinates; equal, up to sign, to the generator P of $U(1)_E$ of the main registry	RS1 6	new
θ_{12}	—	the angle around a vortex line that lies along $x3$, in the plane of $x1$ and $x2$ (with the index 12; θ with other indices and the scattering angle keep the meanings of the main project)	RS1 6	new
`M` (main registry)	`M_gap`	the 8×8 gap matrix of the main registry; `M_gap` is its spelling in the statements of the programs of the revision	RS1 5	main
w_z	`w_z`	a constant spinor of a fermion zero mode on a vortex line	RS1 6	new

D8 Tags and identifiers

symbol	in programs and data files	meaning	defined in	origin
1644d, 1671d	`1644d`, `1671d`	tags of the deflating 4+4 and 7+1 spacetimes and of every object defined on them (the main project's tags 1644, 1671 with the letter d)	DMD 0.1	new
`44d-..`, `71d-..`, `def-..`	—	identifiers of the machine checks of the revision: on the deflating 4+4 spacetime, on the deflating 7+1 spacetime, and for results that hold on both (`44-..`, `71-..` without d are the checks of the main project, executed again)	DMD Appendix A	new
RP0 to RP9, RS1 to RS4, DMD	—	the provenances and supplements of the revision and the document "dirac matrices deflating" (P0 to P11, S1 to S4, DM are the documents of the main project)	README	new
A1, A2, H2, HV	—	A1, A2: the two assumptions of the cosmological part (ordinary space expands; gravity is the eight-dimensional Einstein theory). H2: the author's hypothesis of the main project that the superfluid state is the Higgs field. HV: the different assumption that the vacuum value of the Vorton-Scalar sets the electroweak scale	RP0 3; RS2 1	new

Letters that the revision deliberately does not use, because the main registry or the notebooks use them for other objects: ε and ϵ (an energy density; a transformation parameter), ν and N as free

symbols (the number of internal states; a density), C (a coupling), z (a redshift), ξ (a single-particle energy), σ without index (a self-energy), τ without index (the optical depth), Λ without index (a cutoff), and the names `h0`, `E0`, `mu`, `nu`, `eps`, `Q`, `M`, `S`, `u`, `w0`. Where a first draft used one of them, it was replaced before or after the first commit; the provenance RP9 lists each case (items R-E16, R-E19, R-E20, R-E22).

4. The output of the checks

The following is the file `data/symbol_checks_deflation.txt`, written by `code/scripts/check_symbols.py`:

```
PROJECT spacetimes, Revision_Deflation: checks of the symbols of the revision
generated by: code/scripts/check_symbols.py
5 of 6 checks passed

[PASS] RS1  the 114 names in programs and data files of the 61 entries of the registry of the
 ↪  revision are pairwise different
[PASS] RS2  no name and no symbol that the revision introduces is a name or a symbol of the main
 ↪  registry (78 names of the main registry compared)
[PASS] RS3  none of the 111 names that the revision introduces is a symbol of one of the author's
 ↪  notebooks (1921 symbols compared); in particular g55, g66, g77 do not occur in any notebook
 ↪  (the notebooks use g44 and g71 as names of whole metrics)
[FAIL] RS4  in the 24 documents of the revision the rates g55, g66, g77 are written upright 272
 ↪  times and never as a letter with a subscript; each of the 22 paragraphs that contain a metric
 ↪  component g_{55}, g_{66} or g_{77} says that it is a component of the metric; \varkappa does
 ↪  not occur; rate not upright: provenance/RP1_master_symbols_deflation.md: ... confusion was
 ↪  possible: g55 and $g_{5... | docs_src/RP1_master_symbols_deflation.src.md: ... confusion was
 ↪  possible: g55 and $g_{5...; component without explanation:
 ↪  provenance/RP1_master_symbols_deflation.md: # 2. The one place where confusion was possible:
 ↪  g55 and $g_{55}$ | docs_src/RP1_master_symbols_deflation.src.md: # 2. The one place where
 ↪  confusion was possible: g55 and $g_{55}$
[PASS] RS5  the 13 retired names (tau, kappa_t, eps, Q_over_eps, Omega0, S2, mu and nu as orders
 ↪  of Bessel functions, A, Xi without index, m_b, w0, and omega_c for the shell) do not occur in
 ↪  the 61 data files that the routines of the revision write
[PASS] RS6  each of the 361 columns of the 53 data files of the revision is a registered name of
 ↪  the revision, a registered name or a column of the main project, or one of the 190 descriptive
 ↪  words listed in the script
```

5. How to repeat the proof

```
cd "/Users/nsh/Developer/github/spacetimes/Revision_Deflation"
python3 code/scripts/write_symbol_registry.py     # writes the registry from its source
python3 code/scripts/check_symbols.py             # runs the six checks; exit status 0 if all pass
```

The second command reads the registry of the revision, the MASTER registry of the main project, the list of the symbols of the notebooks (`../provenance/generated/notebook_symbols.csv`), every document of the revision and every data file. It needs only Python 3.